Lexical Databases for Semantic Aggregation: WordNet, VerbNet, FrameNet

NLP Suite TIPS

What is semantic aggregation?

Semantic aggregation means grouping words into higher-level meaning categories - and the reverse, expanding a category into its member words. The NLP Suite's Semantic Aggregation GUI does both, against three complementary lexical resources (WordNet and FrameNet describe themselves as lexical databases, VerbNet as a verb lexicon), chosen from the Knowledge base dropdown (WordNet, VerbNet, FrameNet, or * = all applicable):

- Zoom OUT/UP (aggregate): given a list of words, find the categories they belong to (e.g., run, walk, flee -> a 'motion' category).
- Zoom IN/DOWN (disaggregate): given a category, list its member words (e.g., 'person' -> police, woman, senator...).

The three lexical databases

WordNet (Miller, Princeton)

WordNet is a large lexical database of English in which nouns, verbs, adjectives and adverbs are grouped into synsets - sets of cognitive synonyms, each expressing one concept - interlinked in a hierarchy by conceptual and lexical relations. Unlike a thesaurus, WordNet links specific word senses (so neighbours are sense-disambiguated) and labels the relations between them.

More than 80% of WordNet entries are collocations (multi-word expressions such as coming out, stand in line, customs duty). The NLP Suite uses the noun and verb synsets only (it ignores adjectives, adverbs and function words).

Because WordNet is hierarchical, aggregation climbs UP via hypernymy (X is a kind of Y) and holonymy (X is a part of Y), while disaggregation descends via hyponymy and meronymy. There are 25 top-level NOUN synsets and 15 top-level VERB synsets.

25 top NOUN synsets: act, animal, artifact, attribute, body, cognition, communication, event, feeling, food, group, location, motive, object, person, phenomenon, plant, possession, process, quantity, relation, shape, state, substance, time.

15 top VERB synsets: body, change, cognition, communication, competition, consumption, contact, creation, emotion, motion, perception, possession, social, stative, weather.

Engine. Earlier releases used a Java tool built on the MIT JWI (Java WordNet Interface); the current NLP Suite reimplements WordNet access in Python through NLTK (no Java required). A search from the noun synset 'person' yields some 19,000 proper and improper names of individuals, groups and organizations (saved as social-actor-list.csv in the lib subdirectory); 'animal' yields over 14,000. Because proper nouns have an upper-case initial and improper nouns a lower-case initial, the two can be separated by case (see the Style Analysis GUI's 'Words with capital initial').

Aggregation levels. WordNet is a linked graph, not a simple tree, so a synset can sit at different depths along different paths (e.g., 'anger' is reachable as feeling -> emotion -> anger, but also via state -> ... -> anger). Historically the Suite could only aggregate all the way UP to the 25/15 top synsets. It can now also aggregate to lower-level ANCHOR synsets that YOU choose (typed in the category field): each word climbs to the nearest of your anchors - e.g., anchors person, artifact give 'violence against people vs things'.

VerbNet (based on Levin; built by Palmer group)

Beth Levin, in English Verb Classes and Alternations (1993), made the foundational observation behind VerbNet: verbs that share the same set of syntactic ALTERNATIONS (the grammatical frames they can and cannot appear in) tend to share components of MEANING. For instance, break, shatter and crack all allow the causative/inchoative alternation ('John broke the vase' / 'the vase broke') and form a coherent semantic class. Levin used this syntactic behaviour to sort about 3,000 English verbs into classes.

VerbNet (Kipper Schuler, Palmer et al., University of Colorado) turns Levin's classes into a computational lexicon: each class (e.g., murder-42.1) lists its member verbs together with thematic roles (Agent, Patient, ...) and the syntactic frames in which they occur. VerbNet covers VERBS only.

FrameNet (Fillmore, Berkeley)

FrameNet is built on Charles Fillmore's frame semantics: a word's meaning can only be understood against a FRAME - a schematic situation with participant roles called frame elements. The verb buy evokes a Commerce frame with a Buyer, Seller, Goods and Money. FrameNet (Baker, Fillmore & Lowe, 1998) catalogues these frames and the lexical units (words) that evoke them. It covers NOUNS and VERBS - including event nouns such as assassination.

	WordNet	VerbNet	FrameNet
organizing unit	synset	verb class	frame
how many	~117,000 synsets (25/15 top)	429 classes	1,221 frames

parts of speech	noun, verb	verb only	noun, verb
grouping principle	conceptual synonymy / hierarchy	Levin syntactic alternations	Fillmore situational frames

Are these categories hierarchical? (can you zoom down to finer ones?)

The three are organized very differently, which matters when you want to drill from a broad category into more refined ones:

	Structure	Zoom down to finer categories?
WordNet	Deep IS-A hierarchy (hypernym -> hyponym)	Yes, many levels - e.g. person is 7+ levels deep on the noun side; you can keep descending to ever-finer synsets. This is WordNet's defining structure and what Zoom IN/DOWN exploits.
VerbNet	Shallow: 237 top classes + 192 subclasses	One to two levels - e.g. admire-31.2 -> subclass admire-31.2-1. Subclasses keep the parent's core meaning but add syntactic/role restrictions. (The decimal numbering, e.g. 42.1/42.2, is sibling refinement; the -1 suffix is the actual subclass step.)
FrameNet	Network (DAG) of typed frame relations	Partly - the Inheritance relation gives narrower children (e.g. Killing -> Execution), but a frame can have several parents and other relation types (Using, Causative_of, Subframe, Precedes), so it is a graph, not a clean tree.

In short: WordNet is genuinely, deeply hierarchical; VerbNet has a thin class->subclass hierarchy; FrameNet is a relational network in which 'inherits-from' is only one of several links.

Disambiguation: why the SRL route matters (Palmer / SemLink)

A bare verb is polysemous - hang belongs to several VerbNet classes, and lynch may be missing from a class entirely. Looking up a lemma in isolation therefore gives noisy results (the GUI's list-based VerbNet/FrameNet look-ups take the first / most basic class or frame and warn about this).

The accurate path starts from a disambiguated predicate SENSE, obtained through Semantic Role Labeling (SRL). This is made possible by Martha Palmer's work:

- PropBank (Palmer, Gildea & Kingsbury, 2005) annotates each verb sense with its semantic roles - the layer an SRL system predicts.
- SemLink (Palmer, 2009) is the bridge mapping PropBank sense <-> VerbNet class <-> FrameNet frame.

So once SRL tags a predicate with its PropBank sense, SemLink resolves the CORRECT VerbNet class and FrameNet frame for that occurrence (e.g., murder.01 -> VerbNet 42.1 -> FrameNet Killing) - no guessing from the bare lemma. VerbNet/FrameNet aggregation is therefore most reliable when fed from SRL output rather than a raw word list.

Using the GUI

- Knowledge base - choose WordNet, VerbNet, FrameNet, or * (every applicable resource; VerbNet is skipped for NOUN).
- Lexical category - NOUN or VERB. VerbNet is VERB-only; WordNet and FrameNet handle both.

Zoom IN/DOWN - build a word list from a category

Pick (or type) the category in the category field - for WordNet you can also pick from the top-synset dropdown:

- WordNet: a synset (e.g., person, or a lower-level ethnic group) -> its hyponyms/meronyms.
- VerbNet: a class (murder-42.1) or a member verb (murder) -> the class member verbs.
- FrameNet: a frame (Killing) or a word -> the frame lexical units.

Zoom OUT/UP - aggregate a word list into categories

Select the lemma csv:

- WordNet: climbs to the 25 noun / 15 verb top synsets, or to your own anchor synsets (lower-level aggregation).
- VerbNet: each verb -> its class.
- FrameNet: each word -> its frame.

Input / Output

Zoom IN/DOWN needs no corpus (just the category). Zoom OUT/UP needs a lemma csv - e.g., exported from a CoNLL table via 'Extract nouns & verbs from CoNLL'. Output is a word/category csv plus a frequency csv and chart. To aggregate a raw corpus, first parse it with your default parser (Parsers and annotators - see the GUIs dropdown), then analyse the resulting CoNLL table.

When to use which

- WordNet - conceptual, hierarchical grouping of nouns or verbs (social actors from person; motion verbs).
- VerbNet - verb classes grounded in syntactic behaviour (Levin); good for argument-structure / role analysis.
- FrameNet - situational frames and their roles; good for 'what kind of event/scene' analysis, and the only one with rich event-noun coverage.

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